Document 3 — Dataset Acquisition Guide

SIH26166 · Exactly what to download, from where, and in what order

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1. The five sources you actually need, and nothing else

Your PS names these sources directly. Don't go looking for alternatives — every one of these has a specific, non-interchangeable role.

#	Source	Role	URL
1	Chandrayaan Map Browse	Browse/footprint search for your **source** images	`chmapbrowse.issdc.gov.in`
2	ISSDC PRADAN	Full-resolution, PDS4-labelled raw **source** archive	`pradan.issdc.gov.in`
3	LROC NAC	Primary **reference** imagery	`lroc.im-ldi.com`
4	LROC QuickMap	Interactive coverage/coordinate check for the reference	`quickmap.lroc.im-ldi.com`
5	SELENE (Kaguya), JAXA	Secondary **reference** imagery	JAXA lunar data archive (search "SELENE Kaguya data archive" — JAXA's portal structure changes; use their current data-archive index rather than a bookmarked deep link)

Plus two supporting datasets you'll need but that aren't named as portals in the PS:

#	Source	Role
6	A lunar DEM (LOLA, or the combined SLDEM2015 LOLA+Kaguya-TC mosaic)	Ground-truth elevation for synthetic sun-angle pair generation, ray–surface intersection, and your Fourier-surface terrain rendering
7	A lunar crater catalog (e.g. the LPI/USGS Lunar Crater Database, or Robbins 2019 global crater catalog)	External, geometry-based anchor points for the self-consistency validation check (your own Log 04, Open 5)

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2. Step-by-step: registering and pulling data, in the order to actually do it

Step 1 — Register on PRADAN, today, before anything else

This is explicitly the longest lead-time item in your own workplan.

1. Go to `pradan.issdc.gov.in`.
2. Create an account with institutional email if available (approval can be faster/more reliable for `.ac.in`/university-affiliated addresses).
3. Fill the registration form; note that **approval is manual and not instant** — this is why it goes first, on day one, owned by one team member who does nothing else that day.
4. While waiting, proceed to Step 2 using only the browse portal (no registration required for browsing).

Step 2 — Find a candidate scene pair on Chandrayaan Map Browse

1. Go to `chmapbrowse.issdc.gov.in`.
2. Use the footprint/region search to browse OHRC, TMC-2, and IIRS coverage. Prioritize a **well-documented, named crater** (easier to cross-check visually and against a crater catalog later) rather than an arbitrary coordinate.
3. Note down: instrument, orbit/scene ID, approximate lat/lon bounding box, acquisition date (needed later for sun-angle/ephemeris computation).
4. Pick 2–3 candidates, not just one — some will fail the coverage check in Step 3.

Step 3 — Confirm reference coverage on LROC QuickMap before committing

1. Go to `quickmap.lroc.im-ldi.com`.
2. Enter the same lat/lon bounding box from Step 2.
3. Confirm NAC imagery actually exists over that footprint at a usable resolution/incidence angle — **do this before downloading anything full-resolution**. OHRC's footprint is tiny (~12×3 km), so a near-miss in coordinates can mean zero actual overlap.
4. If NAC coverage is thin or absent, check SELENE coverage for the same region as a fallback (Source 5) — treat it as a genuinely separate case, not a substitute with identical properties (different resolution, different sun-angle history).
5. Repeat for your 2–3 candidates; keep the one with the cleanest confirmed overlap.

Step 4 — Pull the full-resolution source data via PRADAN (once approved)

1. Log in to `pradan.issdc.gov.in`.
2. Search by the orbit/scene ID noted in Step 2.
3. Download the PDS4-labelled product: you'll get an `.xml` label file paired with an `.img` raw data file (OHRC/TMC-2), or a hyperspectral cube for IIRS.
4. Verify the download against the label's stated checksum/size before assuming it's good — a truncated download of a binary `.img` file will not necessarily throw an obvious error when opened.

Step 5 — Pull the reference product

1. From LROC NAC (`lroc.im-ldi.com`): locate the exact NAC product ID confirmed in Step 3, download it (also a PDS-family format).
2. If using SELENE: locate the equivalent product on JAXA's archive for the same footprint.

Step 6 — Get a DEM for your region

- **LOLA gridded data products** (NASA PDS Geosciences Node) give you a lunar-wide altimetry DEM at various resolutions (e.g. 118 m/px global, higher-resolution regional mosaics where available).
- **SLDEM2015** (a combined LOLA + Kaguya Terrain Camera DEM, higher resolution than LOLA alone in the ±60° latitude band) is the better choice if your chosen crater falls in that band — check the PDS Geosciences Node's SLDEM2015 archive page for coverage and download instructions.
- Crop to your scene's footprint immediately after download; don't carry a lunar-wide DEM through your whole pipeline.

Step 7 — Get the crater catalog subset for your region

- Pull the relevant regional subset from the LPI/USGS Lunar Crater Database or the Robbins (2019) global lunar crater catalog (search the current hosting location — these are typically distributed as shapefiles/CSV via the USGS Astrogeology Science Center or the PDS Annex).
- Filter to craters whose rims fall inside your chosen scene's footprint; these become your external, geometry-based validation anchors (Log 04, Open 5).

Step 8 — Verify the ISRO SAC benchmark citation before it goes anywhere near a slide

- The pitch's central "receipt" is the 2025 ISRO Space Applications Centre paper (Makharia et al.) benchmarking classical vs. deep-learning feature matching on Chandrayaan-2 imagery.
- **Verify the exact title, author list, venue, and DOI yourself** before citing it — per your own AI-boundary document, a fabricated-sounding-plausible citation is a real risk category, and this is the one citation every judge in the room is most likely to ask you to produce on the spot.

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3. File formats you will actually open, and with what

Format	Instrument	Read with
PDS4 `.xml` label + `.img` raw array	OHRC, TMC-2	`rasterio`, `GDAL`, or `pds4_tools`
Hyperspectral cube (explicit BSQ/BIP/BIL band order)	IIRS	`spectral` package, or manual `numpy` reads respecting the label's stated interleave

Format	Instrument	Read with
PDS-family raster	LRO NAC	`rasterio/GDAL` (LRO products are widely GDAL-compatible)
PDS-family raster	SELENE/Kaguya	`rasterio/GDAL`; verify projection/datum metadata separately (Log 07 geodesy gap)
GeoTIFF/IMG DEM	LOLA / SLDEM2015	`rasterio`
Shapefile/CSV	Crater catalog	`geopandas` / `pandas`

None of these open in a standard image viewer. Build your ingestion layer against GDAL/rasterio from day one, generically across all three source instruments, per your own workplan's Log 02 checklist.

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4. A ready-to-run data folder structure

```
data/
|-- raw/
|   |-- ohrc/<scene_id>/{*.xml,*.img}
|   |-- tmc2/<scene_id>/{*.xml,*.img}
|   `-- iirs/<scene_id>/{*.xml,*.cube}
|-- reference/
|   |-- lro_nac/<product_id>/*
|   `-- selene/<product_id>/*
|-- dem/
|   `-- <region_name>_sldem2015.tif
|-- crater_catalog/
|   `-- <region_name>_craters.csv
`-- processed/
    |-- synthetic_pairs/
    `-- metrics/
```

Keep `raw/` and `reference/` read-only once downloaded (checksum-verified); do all cropping/reprojection/synthetic-pair generation into `processed/`, never mutating the archive originals in place.